Gargling for Oral Hygiene and the Development of Fever in Childhood: A Population Study in Japan

Background Fever is one of the most common symptoms among children and is usually caused by respiratory infections. Although Japanese health authorities have long recommended gargling to prevent respiratory infections, its effectiveness among children is not clear. Methods The children in this observational study were enrolled from 145 nursery schools in Fukuoka City, Japan. Children in the exposure group were instructed to gargle at least once a day. The endpoints of this study were incidence of fever during the daytime and incidence of sickness absence. Differences among gargling agents for each endpoint were also analyzed. Results A total of 19 595 children aged 2 to 6 years were observed for 20 days (391 900 person-days). In multivariate logistic regression, the overall odds ratio (OR) for fever onset in the gargling group was significantly lower (OR = 0.68). In age-stratified analysis, ORs were significantly lower at age 2 (OR = 0.67), 4 (OR = 0.46), and 5 (OR = 0.41) years. Regarding sickness absence, the overall OR was 0.92 (not significant) in the gargling group. In age-stratified analysis, ORs were significantly lower at age 4 (OR = 0.68), 5 (OR = 0.59), and 6 (OR = 0.63) years. In subgroup analysis, significantly lower ORs for fever onset were observed for children who gargled with green tea (OR = 0.32), functional water (OR = 0.46), or tap water (OR = 0.70). However, the ORs were not significant for sickness absence. Conclusions Gargling might be effective in preventing febrile diseases in children.